Linux Commands Investigation

Day 1 Investigation Worksheet — detailed responses

pwd

What did I ask?	What is the purpose of the pwd command in Linux?
What did AI tell me?	pwd stands for Print Working Directory. It displays the absolute path of the directory in which the current terminal session is operating.
What did I expect?	A command that displays the current directory's full path.
What command/action did I perform?	pwd
What happened?	The terminal displayed the absolute path of the current working directory.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	pwd helps identify the current working directory and avoids confusion when using relative paths.

ls vs ls -la

What did I ask?	What is the difference between ls and ls -la?
What did AI tell me?	ls lists directory contents. The -l option displays detailed information, while -a includes hidden files and directories whose names usually begin with a dot.
What did I expect?	ls would provide a basic listing, while ls -la would display hidden files and detailed information about directory entries.
What command/action did I perform?	ls; ls -la
What happened?	ls displayed a basic listing. ls -la displayed a detailed listing including hidden files, permissions, ownership, sizes, and modification times.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	ls is useful for quickly viewing contents, while ls -la helps inspect hidden configuration files, permissions, and ownership.

cd

What did I ask?	What is the purpose of the cd command in Linux?
What did AI tell me?	cd stands for Change Directory. It changes the current working directory to a specified location.
What did I expect?	A command used to navigate between directories in the Linux filesystem.
What command/action did I perform?	cd /tmp; pwd; cd ~; pwd
What happened?	Changed to /tmp and verified the location with pwd. Then navigated to the current user's home directory using cd ~ and verified the location again.

Yes / No / Partially (AI correctness)	Yes
What did I learn?	cd is essential for navigation. Absolute paths start from the root directory, while relative paths are interpreted from the current directory. cd ~ navigates to the current user's home directory.

mkdir

What did I ask?	What is the purpose of the mkdir command in Linux?
What did AI tell me?	mkdir stands for Make Directory. It creates directories at the specified path. The -p option can create missing parent directories.
What did I expect?	A command that creates a new directory in the specified location.
What command/action did I perform?	mkdir test-dir; ls -l; mkdir -p parent/child
What happened?	Created test-dir and verified it with ls -l. Used mkdir -p to create a nested directory structure, including missing parent directories.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	mkdir creates directories, and -p is useful for nested directory structures. Choosing an appropriate location and name helps keep files organized.

rm

What did I ask?	What is the purpose of the rm command in Linux?
What did AI tell me?	rm removes files and, with suitable options, directories. The -r option enables recursive removal, while -f forces removal and suppresses certain prompts and errors.
What did I expect?	A command used to delete unwanted files and directories.
What command/action did I perform?	touch test.txt; ls -l test.txt; rm test.txt; ls -l test.txt
What happened?	Created a test file, verified that it existed, removed it, and checked again. The final listing indicated that the file no longer existed.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	rm deletes files, and recursive options can remove directory trees. Deletion can be difficult to reverse, so the target path and options must be checked carefully.

Challenge: rm -rf

What did I ask?	What happens if I execute rm -rf, and why is it dangerous?
What did AI tell me?	The -r option enables recursive deletion, and -f forces removal while suppressing certain prompts and errors. With a valid target, it can remove files and directories recursively without asking for confirmation.
What did I expect?	A command that can delete entire directory structures and their contents, potentially causing serious data loss if used incorrectly.

What command/action did I perform?	Investigated the command's behavior and options. No destructive command needs to be executed to understand its effects.
What happened?	Learned that the command can recursively delete a specified target and its contents. Its impact depends on the target path, permissions, and system protections.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	rm -rf is dangerous because a mistake in the target path can cause significant data loss. Verify paths carefully, avoid unnecessary elevated privileges, and never execute unfamiliar destructive commands blindly.

Final Learning

The investigation helped me understand how to navigate directories, inspect files, create folders, and remove files. I also learned to be especially careful with destructive commands such as rm -rf.